

Software Requirements Specification (SRS)

Project: Hospital Management System (HMS)

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1. Introduction

1.1 Purpose

This document is the Software Requirements Specification (SRS) for the Hospital Management System (HMS). It defines the system's functional and non-functional requirements, external interfaces, security objectives and requirements, use-case models, and verification criteria. The HMS supports patient registration and identity management, OPD appointment scheduling, clinical consultations and electronic health records, laboratory test orders and results, inpatient admission and bed management, pharmacy and inventory operations, billing and payment recording, staff and access administration, and operational reporting. This document is intended for developers, testers, system integrators, hospital IT administrators, course instructors, and project evaluators.

1.2 Scope

The Hospital Management System (HMS) shall support outpatient (OPD) and inpatient (IPD) workflows, including patient registration and identity management, appointment scheduling, clinical consultation and electronic health records (EHR), bed/ward management, pharmacy and inventory tracking, billing and payments, and administrative reporting. The system is implemented as a client-server application (C/C++ back end with a relational database) accessed by hospital staff through desktop/kiosk front ends.

Out of scope: This SRS does not cover integration with external national health-data exchanges beyond a defined API stub, deep insurance-claim adjudication logic, medical-device driver development, or hospital financial/ERP accounting beyond bill generation and payment recording.

1.3 Audience

Software Developers, QA Engineers, System Integrators, Hospital IT Administrators, Course Instructors, and Project Evaluators.

1.4 Definitions, Acronyms and Abbreviations

- HMS – Hospital Management System
- EHR – Electronic Health Record
- UHID – Unique Hospital Identification (patient ID)
- OPD – Out-Patient Department
- IPD – In-Patient Department
- UI – User Interface
- API – Application Programming Interface
- DB – Database
- RBAC – Role-Based Access Control
- TLS – Transport Layer Security
- LIS – Laboratory Information System
- SMS – Short Message Service
- PII – Personally Identifiable Information
- RTM – Requirements Traceability Matrix
- NFR – Non-Functional Requirement

2. Overall description

2.1 Product perspective

The HMS is a standalone hospital information system comprising a central application server, a relational database, and multiple client front ends (reception desk, clinical workstation, pharmacy counter, billing counter, admin dashboard, and an optional patient self-service kiosk). It replaces manual/paper-based registers for patient records and appointment booking, and interfaces with peripheral hardware (barcode/RFID scanners, printers) and optional external services (SMS/e-mail gateway, payment gateway, laboratory systems).

2.2 Major product functions (detailed)

- Patient registration and UHID (Unique Hospital ID) management

- OPD appointment scheduling, rescheduling and cancellation
- Clinical consultation recording and Electronic Health Records (EHR)
- Electronic prescription generation
- Laboratory/radiology test ordering and results recording
- In-patient admission, bed/ward management and discharge summary generation
- Billing, payments and insurance/discount handling
- Pharmacy inventory management, dispensing and expiry tracking
- Staff and role/access management
- Operational and financial reporting

2.3 User roles and characteristics (expanded)

Patient: Books/cancels appointments, views own records and bills; assumed to have basic digital literacy.

Receptionist / Front-desk Staff: Registers patients, schedules appointments, generates bills; needs a fast, low-training UI for high daily transaction volume.

Doctor: Records diagnoses and vitals, writes prescriptions, orders tests, reviews patient history; needs quick access to EHR during consultations.

Nurse: Records vitals, administers treatment, updates ward/bed status for admitted patients.

Pharmacist: Dispenses medicines against prescriptions, manages stock levels and expiry alerts.

Lab Technician: Records laboratory/radiology test results against test orders.

Billing Clerk: Verifies charges, records payments, issues receipts and applies insurance/discount schemes.

Hospital Administrator: Manages staff accounts and roles, views operational reports, configures system parameters.

2.4 Operating environment

Server: Linux/Windows host running the C/C++ application server and a relational database (e.g., MySQL/PostgreSQL). Clients: desktop PCs at reception, clinical, pharmacy and billing counters, plus an optional touch-screen kiosk. Network: hospital LAN with secure remote access for administrators. Power: UPS-backed for server and critical counters. The system shall be deployable within a single hospital site with provision for future multi-branch deployment.

2.5 Constraints

- Core back-end services shall be implemented in C/C++.
- Patient health data handling shall follow applicable data-protection and healthcare-privacy regulations.
- The system shall use only approved cryptographic libraries for data protection.
- Hardware peripherals (scanners, printers) shall be accessed via vendor-supplied APIs/drivers.
- The system shall operate on the hospital's existing LAN infrastructure without requiring new external connectivity for core functions.

3. External interface requirements

3.1 User interfaces

Primary UI: desktop application screens for reception, clinical, pharmacy, billing and admin roles, with a menu-driven and form-based layout. An optional touch-enabled kiosk UI supports patient self-check-in. Accessibility: high-contrast display mode and keyboard-navigable forms for staff with visual constraints.

3.2 Hardware interfaces

- Barcode / RFID scanner for patient wristband and UHID lookup

- Biometric fingerprint reader for staff login (optional, configurable)
- Receipt / bill printer at billing and pharmacy counters
- Standard keyboard, mouse and touch-screen input devices at kiosks

3.3 Software interfaces

- Relational Database Management System (RDBMS) for persistent storage of patient, clinical, billing and inventory data
- Laboratory Information System (LIS) interface for transmitting test orders and receiving results
- Payment Gateway / UPI interface for electronic payment processing
- SMS / E-mail Gateway API for appointment confirmations and reminders
- Optional national health-data exchange API stub for future interoperability

3.4 Communications

- TLS 1.2 or higher shall be enforced for all network communication between client applications and the application server, between the application server and external services, and between the application server and the database. Connections that cannot negotiate TLS 1.2 or higher shall be rejected.
- REST/JSON APIs shall be used between client front ends and the application server.
- A retry/back-off strategy shall be applied for calls to external services (SMS gateway, payment gateway, LIS).
- Database connections shall use TLS 1.2 or higher. The selected database and deployment configuration shall be verified to support encrypted database connections; plaintext database connections shall not be permitted.

4. System features (detailed)

Each requirement below includes its acceptance criteria and a reference test case. IDs follow HMS-F-###.

4.1 Patient Registration & Identity Management

Description: Register new patients, assign a Unique Hospital ID (UHID), and authenticate staff logins.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-001	The system shall allow front-desk staff to register a new patient by capturing demographic details (name, DOB, gender, contact, address) and assign a unique UHID.	Functional	High	Business / Reception	AC-HMS-F-001: A new UHID is generated and the patient record is retrievable by that UHID. Test: TC-Reg-01	None
HMS-F-002	The system shall allow staff to search and retrieve an existing patient record by UHID, name, or phone number, supporting partial matches.	Functional	High	Reception / Clinical	AC-HMS-F-002: Searching by any of the three keys returns the correct patient within 2 seconds. Test: TC-Reg-02	Requires indexed DB search
HMS-F-003	The system shall maintain a longitudinal Electronic Health Record (EHR) per patient, including visit history, diagnoses, prescriptions and lab results.	Functional	High	Clinical / Compliance	AC-HMS-F-003: All prior visits for a UHID are listed in chronological order. Test: TC-Reg-03	Depends on HMS-F-009, HMS-F-010, HMS-F-011
HMS-F-004	The system shall authenticate staff logins via username/password and lock an account after 5 consecutive failed attempts, logging an audit event.	Functional	High	Security	AC-HMS-F-004: After 5 failures the account is locked and an audit entry is recorded. Test: TC-Auth-01	See HMS-SR-006 for failed-login lockout; see HMS-SR-002 for role-based access control.

4.2 Appointment Scheduling

Description: Book, reschedule and cancel OPD appointments with real-time slot availability.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-005	The system shall allow a patient or receptionist to book an OPD appointment with a chosen doctor, department, date and time slot, subject to availability.	Functional	High	Patient / Reception	AC-HMS-F-005: Booking succeeds only for an available slot and the slot is then marked unavailable. Test: TC-Appt-01	Requires doctor schedule config
HMS-F-006	The system shall prevent double-booking of a doctor's time slot and display real-time slot availability to the user.	Functional	High	Operations	AC-HMS-F-006: Concurrent booking attempts for the same slot result in exactly one success. Test: TC-Appt-02	Concurrency control required
HMS-F-007	The system shall allow an appointment to be rescheduled or cancelled up to a configurable cutoff time, updating slot availability accordingly.	Functional	Medium	Patient / Reception	AC-HMS-F-007: Cancelling releases the slot; rescheduling moves the booking without creating a duplicate. Test: TC-Appt-03	Cutoff time configurable by admin
HMS-F-008	The system shall send appointment confirmation and reminder notifications to the patient via SMS/e-mail.	Functional	Medium	Patient Experience	AC-HMS-F-008: A confirmation is sent within 1 minute of booking and a reminder 24 hours prior. Test: TC-Appt-04	Requires SMS/e-mail gateway

4.3 Consultation & Clinical Records

Description: Capture clinical notes, prescriptions and diagnostic test orders/results against a patient visit.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-009	The system shall allow a doctor to record diagnosis, vitals and clinical notes against a specific patient visit.	Functional	High	Clinical	AC-HMS-F-009: A saved consultation note is linked to the correct visit and UHID. Test: TC-Clin-01	None
HMS-F-010	The system shall allow a doctor to create an electronic prescription specifying drug name, dosage, frequency and duration.	Functional	High	Clinical / Pharmacy	AC-HMS-F-010: The prescription is viewable by pharmacy staff against the correct UHID. Test: TC-Clin-02	Feeds HMS-F-017
HMS-F-011	The system shall allow authorized staff to order laboratory/radiology tests and record results linked to the patient visit.	Functional	High	Clinical / Lab	AC-HMS-F-011: A test order status transitions Ordered → Completed with results attached. Test: TC-Clin-03	Requires LIS interface

4.4 In-Patient (IPD) & Bed Management

Description: Manage bed/ward allocation for admitted patients and generate discharge summaries.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-012	The system shall track bed/ward availability and allow admission of a	Functional	High	Operations / Nursing	AC-HMS-F-012: An occupied bed cannot be assigned to a second patient. Test: TC-IPD-01	None

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
	patient to a specific bed with an admission record.					
HMS-F-013	The system shall generate a discharge summary consolidating diagnosis, treatment given and follow-up instructions upon patient discharge.	Functional	Medium	Clinical	AC-HMS-F-013: The discharge summary is generated and printable within 1 minute of discharge confirmation. Test: TC-IPD-02	Depends on HMS-F-009

4.5 Billing & Payments

Description: Consolidate charges, record payments and apply discount/insurance schemes.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-014	The system shall generate a consolidated bill covering consultation, tests, medication and room charges for a visit or admission.	Functional	High	Billing / Finance	AC-HMS-F-014: The bill total equals the sum of all itemized charges for the visit. Test: TC-Bill-01	Depends on 4.1–4.4 data
HMS-F-015	The system shall support recording of payments via cash, card and UPI; insurance/discount adjustments are recorded separately, and shall print or e-mail a receipt.	Functional	High	Billing / Patient	AC-HMS-F-015: A receipt is generated immediately after payment is recorded. Test: TC-Bill-02	Payment gateway required for card/UPI; insurance and discounts are separate coverage/adjustment records.
HMS-F-016	The system shall apply configured insurance or discount schemes to an eligible bill before final invoice generation.	Functional	Medium	Billing / Business	AC-HMS-F-016: The discounted total matches the configured scheme percentage/amount. Test: TC-Bill-03	Scheme rules configurable by admin

4.6 Pharmacy & Inventory

Description: Track medicine stock, dispensing and expiry.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-017	The system shall decrement pharmacy stock automatically when a prescribed medicine is dispensed and flag items below a reorder threshold.	Functional	High	Pharmacy / Operations	AC-HMS-F-017: Stock count decreases by the dispensed quantity and a low-stock flag appears when below threshold. Test: TC-Pharm-01	Depends on HMS-F-010
HMS-F-018	The system shall alert pharmacy staff of medicines nearing expiry within a configurable window (e.g., 60 days).	Functional	Medium	Pharmacy / Compliance	AC-HMS-F-018: An expiry alert list is generated daily for batches expiring within the configured window. Test: TC-Pharm-02	Window configurable by admin

4.7 Administration & Reporting

Description: Manage staff accounts/roles and generate operational reports.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance Criteria / Test Case Ref	Comments / Dependencies
HMS-F-019	The system shall allow an administrator to create, modify and deactivate staff accounts and assign roles/permissions.	Functional	High	Admin / Security	AC-HMS-F-019: A deactivated account can no longer log in; role changes take effect on next login. Test: TC-Admin-01	See HMS-SR-002
HMS-F-020	The system shall generate operational reports (daily OPD footfall, revenue, bed occupancy) for a specified date range, exportable to PDF/CSV.	Functional	Medium	Admin / Management	AC-HMS-F-020: The exported report totals match underlying transaction records for the selected range. Test: TC-Admin-02	None

Summary: 20 functional requirements (HMS-F-001 through HMS-F-020) are specified above, exceeding the required minimum of 15.

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs follow HMS-NF-###.

Req ID	Requirement (shall...)	Category	Priority	Acceptance Criteria / Measurement
HMS-NF-001	90% of appointment-booking transactions shall complete within 3 seconds under normal load (up to 200 concurrent users).	Performance	High	Load test shows P90 $\leq$ 3s at 200 concurrent users. Test: TC-Perf-01
HMS-NF-002	The system shall provide 99.5% availability during hospital operating hours (06:00–23:00), excluding scheduled maintenance windows.	Reliability	High	Uptime monitoring reports $\geq$ 99.5% monthly. Test: Ops report review
HMS-NF-003	All patient health information shall be encrypted at rest (AES-256) and in transit (TLS 1.2+).	Security / Compliance	High	Encryption audit / config review confirms AES-256 at rest and TLS 1.2+ in transit. Test: TC-Sec-01
HMS-NF-004	Front-desk and clinical UI screens shall be operable by new staff after no more than 2 hours of training, and shall support a high-contrast display mode.	Usability / Accessibility	Medium	Usability test with new staff achieves task completion after $\leq$ 2 hours training. Test: TC-UX-01
HMS-NF-005	The system shall be modular (separate patient, appointment, billing and pharmacy modules) such that any single module can be updated/redeployed with total downtime not exceeding 15 minutes. The 15-minute interruption limit applies to the affected module; other modules should remain available where their dependencies permit.	Maintainability	Medium	A module redeploy in staging completes with $\leq$ 15 min service interruption. Test: TC-Maint-01 The 15-minute interruption limit applies to the affected module; other modules should remain available where their dependencies permit.
HMS-NF-006	The appointment and records modules shall support horizontal scaling to accommodate at least 3x current patient load without architectural change. This target is measured in a documented load test against the baseline workload, without violating the response-time target in HMS-NF-001.	Scalability	Medium	Load test at 3x baseline load meets HMS-NF-001 response-time targets after scaling out. Test: TC-Perf-02 This target is measured in a documented load test against the baseline workload, without violating the response-time target in HMS-NF-001.

Summary: 6 non-functional requirements are specified above, exceeding the required minimum of 5.

5.1 Security

5.1.1 Security Objectives

SO-1 (Confidentiality): Ensure that patient health records and personally identifiable information (PII) are accessible only to authorized roles on a strict need-to-know basis.

SO-2 (Integrity & Availability): Ensure that clinical, prescription and billing data cannot be altered without authorization and a recorded audit trail, and remain available to authorized clinical staff throughout hospital operating hours.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance Criteria / Test Case Ref
HMS-SR-001	TLS 1.2+ shall be mandatory for all network connections between client applications, the application server and the database.	Security	High	AC-HMS-SR-001: Connection attempts using TLS < 1.2 or plaintext are rejected. Test: TC-Sec-02
HMS-SR-002	The system shall enforce role-based access control (RBAC) such that a receptionist cannot view clinical notes and a pharmacist cannot view billing details outside their scope.	Security	High	AC-HMS-SR-002: Access attempts outside a role's permitted scope are denied and logged. Test: TC-Sec-03
HMS-SR-003	The system shall maintain an immutable audit log of all create/read/update/delete operations on patient health records, including user ID, timestamp and action.	Security	High	AC-HMS-SR-003: Every CRUD operation on an EHR record produces a corresponding audit-log entry. Test: TC-Sec-04
HMS-SR-004	Passwords shall be stored using a salted one-way hash (e.g., bcrypt/Argon2) and shall never be stored or transmitted in plaintext.	Security	High	AC-HMS-SR-004: Database inspection shows only salted hashes; network capture shows no plaintext passwords. Test: TC-Sec-05
HMS-SR-005	The system shall enforce automatic session timeout after 10 minutes of inactivity on clinical and administrative workstations.	Security	Medium	AC-HMS-SR-005: An idle session is terminated at 10 minutes and requires re-authentication. Test: TC-Sec-06
HMS-SR-006	The system shall temporarily lock a staff account after five consecutive failed authentication attempts. The lockout duration or administrator-assisted recovery process shall be documented, and each lockout event shall be logged. Patient self-service accounts shall use the same or stronger configurable protection against repeated failed login attempts.	Security	High	AC-HMS-SR-006: The sixth consecutive failed attempt is blocked or requires the defined recovery process; a lockout event is recorded. Test: TC-Sec-06

Summary: 2 security objectives and 6 security requirements are specified above.

6. Quality attributes & Acceptance tests

Exit criteria for acceptance: All High-priority functional requirements are implemented and verified, no critical NFR or security-requirement failures remain open, and the RTM (Section 8) shows all associated test cases passed.

Acceptance test suites: Registration & Authentication, Appointment Scheduling, Clinical Consultation, IPD/Bed Management, Billing & Payments, Pharmacy & Inventory, Administration & Reporting, Performance, Security, and Accessibility.

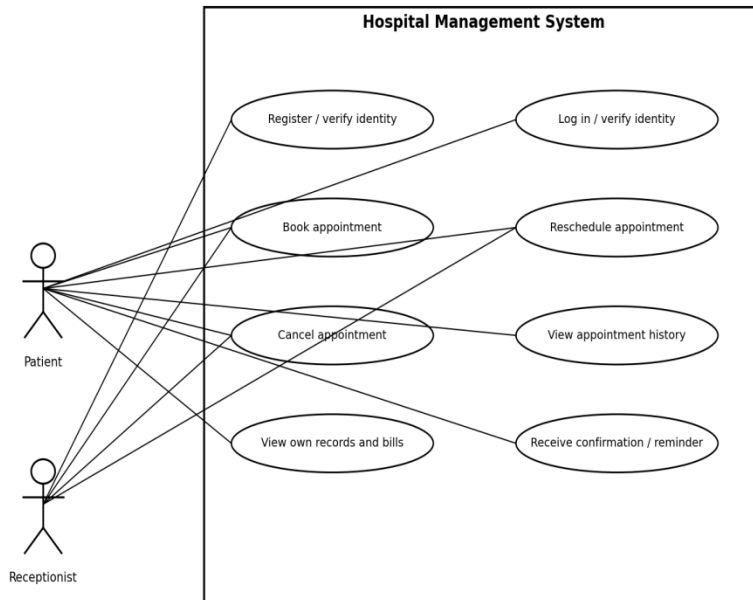
Each functional and non-functional requirement's individual acceptance criterion and test-case reference is documented inline in its requirement table in Sections 4 and 5 above.

7. System models and diagrams

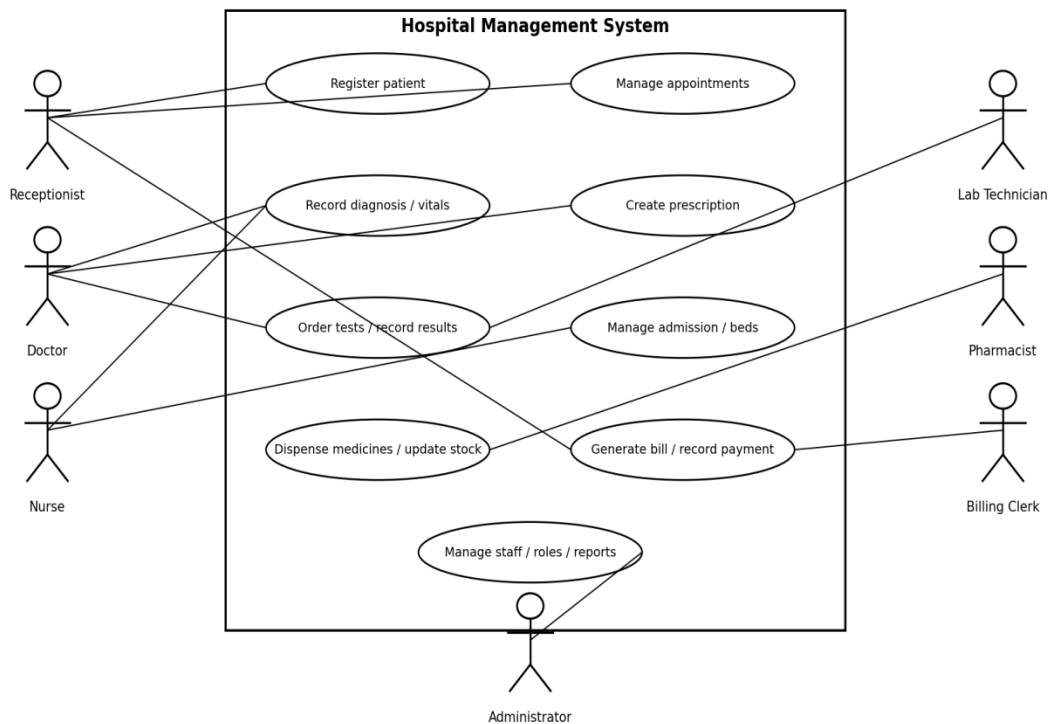
7.1 UML Use-Case Diagrams

Two UML use-case diagrams are provided below: one for patient-facing interactions, and one for clinical and administrative staff interactions. The diagrams are illustrative and show actor-to-use-case associations only; use-case-to-use-case relationships are shown only where a valid UML <<include>> or <<extend>> relationship is intended.

Patient-Facing Use Cases



Clinical and Administrative Staff Use Cases



8. Requirements Traceability Matrix (RTM)

The RTM below traces every functional requirement, non-functional requirement and security requirement to its design/section reference, implementing module, and test case(s). Status: N = Not tested, P = Pass, A = deferred to future release.

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
HMS-F-001	Register patient / assign UHID	4.1 / DS-Reg-01	RegistrationModule	TC-Reg-01	N	
HMS-F-002	Search patient record	4.1 / DS-Reg-02	RegistrationModule	TC-Reg-02	N	
HMS-F-003	Maintain EHR	4.1 / DS-EHR-01	EHRModule	TC-Reg-03	N	
HMS-F-004	Staff login / logout	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	See HMS-SR-006 for failed-login lockout; see HMS-SR-002 for role-based access control.
HMS-F-005	Book OPD appointment	4.2 / DS-Appt-01	AppointmentModule	TC-Appt-01	N	
HMS-F-006	Prevent double-booking	4.2 / DS-Appt-02	AppointmentModule	TC-Appt-02	N	
HMS-F-007	Reschedule / cancel appointment	4.2 / DS-Appt-03	AppointmentModule	TC-Appt-03	N	
HMS-F-008	Appointment notifications	4.2 / DS-Appt-04	NotificationModule	TC-Appt-04	N	Requires SMS/e-mail gateway
HMS-F-009	Record diagnosis / vitals	4.3 / DS-Clin-01	ClinicalModule	TC-Clin-01	N	
HMS-F-010	Create prescription	4.3 / DS-Clin-02	ClinicalModule	TC-Clin-02	N	
HMS-F-011	Order / record lab tests	4.3 / DS-Clin-03	LabModule	TC-Clin-03	N	Requires LIS interface
HMS-F-012	Bed admission	4.4 / DS-IPD-01	IPDModule	TC-IPD-01	N	
HMS-F-013	Discharge summary	4.4 / DS-IPD-02	IPDModule	TC-IPD-02	N	
HMS-F-014	Consolidated billing	4.5 / DS-Bill-01	BillingModule	TC-Bill-01	N	
HMS-F-015	Record payment / receipt	4.5 / DS-Bill-02	BillingModule	TC-Bill-02	N	Payment gateway required for card/UPI; insurance and discounts are separate coverage/adjustment records.
HMS-F-016	Apply discount / insurance	4.5 / DS-Bill-03	BillingModule	TC-Bill-03	N	
HMS-F-017	Dispense & decrement stock	4.6 / DS-Pharm-01	PharmacyModule	TC-Pharm-01	N	
HMS-F-018	Expiry alerts	4.6 / DS-Pharm-02	PharmacyModule	TC-Pharm-02	N	
HMS-F-019	Manage staff accounts/roles	4.7 / DS-Admin-01	AdminModule	TC-Admin-01	N	See HMS-SR-002
HMS-F-020	Operational reports	4.7 / DS-Admin-02	ReportingModule	TC-Admin-02	N	

Req ID	Requirement (short)	Section Ref / Design Spec	Module	Test Case(s)	Status (N/P/A)	Comments
HMS-NF-001	Booking response time	5 / DS-Perf-01	AppointmentModule	TC-Perf-01	N	
HMS-NF-002	99.5% availability	5 / DS-Ops-01	Infra / Ops	Ops report review	N	
HMS-NF-003	Encryption at rest/in transit	5 / DS-Sec-01	Platform	TC-Sec-01	N	
HMS-NF-004	Usability / training time	5 / DS-UX-01	UI Layer	TC-UX-01	N	
HMS-NF-005	Modular redeploy downtime The 15-minute interruption limit applies to the affected module; other modules should remain available where their dependencies permit.	5 / DS-Maint-01	Platform	TC-Maint-01	N	
HMS-NF-006	Horizontal scalability 3x This target is measured in a documented load test against the baseline workload, without violating the response-time target in HMS-NF-001.	5 / DS-Perf-02	AppointmentModule / EHRModule	TC-Perf-02	N	
HMS-SR-001	TLS 1.2+ enforced	5.1.2 / DS-Sec-02	Platform	TC-Sec-02	N	
HMS-SR-002	RBAC enforcement	5.1.2 / DS-Sec-03	AuthModule	TC-Sec-03	N	
HMS-SR-003	Immutable audit log	5.1.2 / DS-Sec-04	AuditModule	TC-Sec-04	N	
HMS-SR-004	Salted password hashing	5.1.2 / DS-Sec-05	AuthModule	TC-Sec-05	N	
HMS-SR-005	Session timeout (10 min)	5.1.2 / DS-Sec-06	AuthModule	TC-Sec-06	N	
HMS-SR-006	Failed-login lockout	5.1.2 / DS-Sec-06	AuthenticationModule	TC-Sec-06	N	Added to define the account lockout behavior referenced by HMS-F-004.